

Mitchell C. Elliott

(408) 643-4372 | mitch.elliott@pacbell.net

linkedin.com/in/mitchellelliott18 | github.com/melliott18 | melliott18.github.io

Software Engineer | San Francisco Bay Area | U.S. Citizen

SUMMARY

Software engineer who builds applications and infrastructure with the people who use and operate them in mind. Brings experience across cloud platforms, business software, and systems research, with a focus on reducing manual work and making complex workflows easier to run and maintain. Applies that same practical approach to engineering leadership and technical mentoring, helping teams understand and improve the systems they work with.

SKILLS

Languages: Python, C, C++, Go, Java, JavaScript, TypeScript, Bash, SQL

Application Development: React, Node.js, Flask, FastAPI, REST APIs, gRPC

Systems & Cloud: Linux, AWS (EC2, S3, VPC, IAM, ELB), GCP, Kubernetes, Docker, Ceph

Delivery & Automation: Git, GitHub Actions, GitLab CI/CD, AWS CodeBuild, Amazon ECS, Terraform

Data & Storage: PostgreSQL, Redis, OpenSearch, SQLite, BigQuery, DuckDB, Parquet

Observability: New Relic, NRQL, Prometheus, Grafana, AWS CloudWatch

Testing & Performance: Automated testing, GDB, Valgrind, perf, fio

EXPERIENCE

UCSC Center for Research in Systems and Storage

Jan. 2026 – Aug. 2026

Research Software Engineer, Platform Infrastructure

Santa Cruz, CA

- Led a 3-person engineering team developing and operating shared infrastructure for systems research, coordinating platform architecture, implementation, code review, and experiment support across parallel lab projects
- Built and maintained containerized research environments on NRP Nautilus Kubernetes with Ceph-backed persistent storage, supporting repeatable compute- and data-intensive workloads across multiple users
- Standardized deployment, configuration, data-access, and experiment workflows across Kubernetes, Ceph, BigQuery, DuckDB, and Parquet, reducing environment-specific setup and improving reproducibility
- Developed reusable self-service operational tooling for launching, monitoring, and managing long-running experiments, improving resource utilization while avoiding thousands of dollars in equivalent GCP compute costs

UCSC Center for Research in Systems and Storage

Dec. 2024 – Jan. 2026

Research Software Engineer, Storage Systems

Santa Cruz, CA

- Evaluated storage architectures across device classes and workload patterns, analyzing performance, capacity, energy, and cost tradeoffs to guide infrastructure design decisions
- Designed repeatable trace-driven workflows for comparing storage technologies under realistic demand, standardizing inputs, parameter sweeps, validation, and result review across configurations
- Collaborated with systems researchers to translate open-ended research questions into measurable experiments and synthesize results into storage-architecture recommendations

University of California, Santa Cruz

Jan. 2023 – Dec. 2024

Head Teaching Assistant – CSE 130: Principles of Computer System Design

Santa Cruz, CA

- Built and maintained automated grading and feedback workflows for an upper-division systems-programming course serving 1,000+ students, reducing overall grading time by 50%
- Coordinated course operations and mentored teaching staff across six quarters, standardizing documentation, grading workflows, issue escalation, onboarding, and staff coverage
- Taught clean, modular systems code, reusable libraries, unit/regression/end-to-end tests, edge-case discovery, and harnesses; guided GDB debugging and perf profiling to diagnose bottlenecks, concurrency issues, and memory errors

Perfectly Snug

July 2023 – Sept. 2023

Software Engineer Intern

Remote

- Developed a material requirements planning (MRP) application using Python, Node.js, React, and SQL, integrating inventory, customer orders, bills of materials, and production data into a single internal system
- Built operational dashboards for inventory availability, material requirements, order status, and production progress while automating reconciliation and scheduling workflows, reducing manual planning time by roughly 50%
- Worked directly with manufacturing stakeholders to gather requirements, iterate on operational workflows, debug edge cases, and translate production needs into maintainable internal software

ParkourSC, Inc.

July 2022 – Sept. 2022

DevOps Engineer Intern

Remote

- Operated production cloud and platform infrastructure for a supply-chain application deployed across 100+ Kubernetes clusters, spanning IoT ingestion, Kafka, Spark, Cassandra, Elasticsearch, and web services
- Tuned New Relic alerting policies using NRQL and standardized alert review and escalation with developers, reducing false-positive production alerts by approximately 80% across distributed services
- Automated monitoring workflows around Kubernetes deployments and cluster upgrades, integrating New Relic and Jira to suppress maintenance-related alerts while preserving operational context
- Triaged incidents across application, platform, and cloud infrastructure layers by tracing service dependencies and correlating Kubernetes state, dashboards, metrics, and logs to isolate failures

uLab Systems, Inc.

June 2021 – Sept. 2021

Site Reliability Engineer Intern

San Mateo, CA

- Migrated a production application from Lightsail to EC2 with VPC isolation, Elastic Load Balancing, IAM access controls, and CloudWatch monitoring, improving deployment flexibility and operational control
- Wrote a Python script to copy objects and preserve Access Control Lists (ACLs) between Amazon S3 buckets, automating data migration while retaining object-level permissions
- Integrated Datawiza Access Broker with Okta, implementing centralized authentication and proxy-based access controls for the production application
- Built a CI/CD pipeline with GitHub, AWS CodeBuild, and Amazon ECS for development, validation, and production; documented Terraform provisioning and recovery workflows to reduce manual work and standardize releases

PROJECTS

Workload Trace Analysis Platform (UCSC CRSS)

Jan. 2026 – Aug. 2026

- Built a data-processing platform for multi-month Google cluster and power traces stored in Google Cloud Storage, using BigQuery to select and extract bounded source datasets into repeatable local analysis workflows
- Materialized extracted traces as Parquet and used DuckDB to query, transform, join, and validate reusable datasets without repeatedly scanning the complete remote source
- Created CLI and automated testing workflows to orchestrate dataset selection, BigQuery extraction, Parquet conversion, DuckDB processing, schema validation, and packaging of simulator-ready inputs

CarbonStream: Data Center Carbon TCO Simulator (UCSC CRSS)

Jan. 2024 – Present

- Built a Python trace-driven simulator for evaluating performance, capacity, energy use, and carbon impact across heterogeneous data-center storage architectures
- Modeled workload behavior using device latency, throughput, power, lifespan, capacity, and regional carbon-intensity inputs under changing demand
- Implemented bottleneck detection, service-level-objective checks, configuration search, parameter sweeps, and automated CSV/visualization outputs for reproducible analysis

CogniStore: AI-Powered Data Lifecycle Manager

Sept. 2025 – Present

- Developed an intelligent data-management platform unifying file, block, and object storage behind a common API for lifecycle management across heterogeneous storage services
- Built a modular storage abstraction layer integrating PostgreSQL with pgvector, OpenSearch, and Redis for metadata persistence, semantic indexing, and low-latency retrieval
- Implemented an AI-driven policy engine placing data across storage tiers based on lifecycle, cost, performance, and sustainability goals; the project uses GitHub Actions in the CI/CD process to run automated tests

SimOps Workbench: Simulation Experiment Dashboard

June 2026

- Built a TypeScript and React self-service dashboard around a dynamics simulator, turning one-off runs into repeatable workflows for configuring scenarios, monitoring execution, and comparing telemetry
- Integrated Flask APIs and SQLite-backed experiment history with parameter-sweep and batch workflows, allowing completed experiments to be reopened, reproduced, and compared
- Added interactive telemetry comparisons, JSON/CSV exports, printable reports, and Docker Compose workflows to improve analysis, sharing, and reproducibility

EDUCATION

University of California, Santa Cruz

M.S. Computer Science | Sept. 2022 – Dec. 2024

B.S. Computer Science | Sept. 2018 – June 2022